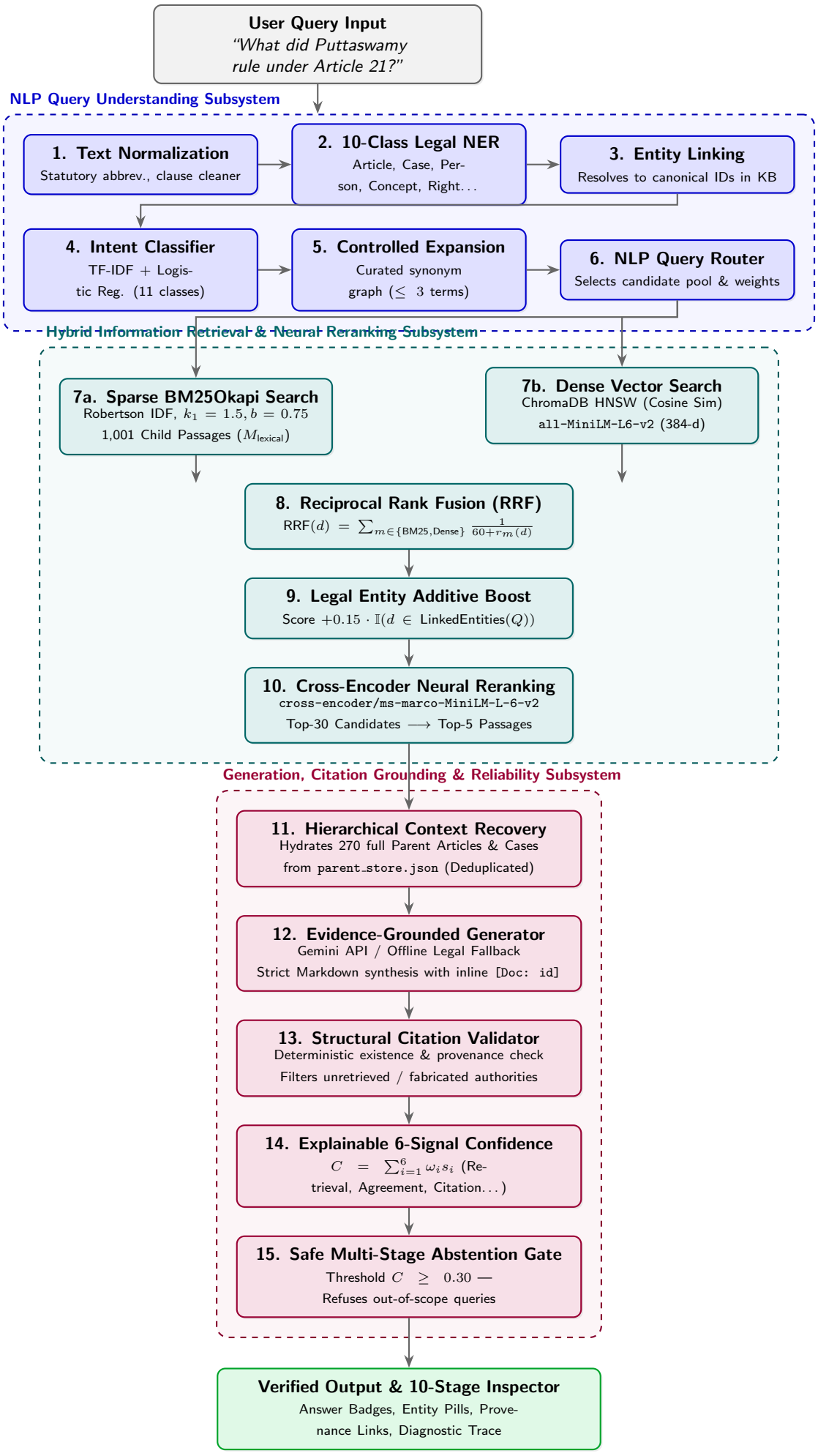

User Query Input
"What did Puttaswamy rule under Article 21?"
NLP Query Understanding Subsystem
1. Text Normalization
Statutory abbrev., clause cleaner
2. 10-Class Legal NER
Article, Case, Person, Concept, Right...
3. Entity Linking
Resolves to canonical IDs in KB
4. Intent Classifier
TF-IDF + Logistic Reg. (11 classes)
5. Controlled Expansion
Curated synonym graph (≤ 3 terms)
6. NLP Query Router
Selects candidate pool & weights
Hybrid Information Retrieval & Neural Reranking Subsystem
7a. Sparse BM25Okapi Search
Robertson IDF, $k_1 = 1.5, b = 0.75$
1,001 Child Passages (M_{lexical})
7b. Dense Vector Search
ChromaDB HNSW (Cosine Sim)
all-MiniLM-L6-v2 (384-d)
8. Reciprocal Rank Fusion (RRF)
$\text{RRF}(d) = \sum_{m \in \{\text{BM25, Dense}\}} \frac{1}{60 + r_m(d)}$
9. Legal Entity Additive Boost
Score $+0.15 \cdot \mathbb{I}(d \in \text{LinkedEntities}(Q))$
10. Cross-Encoder Neural Reranking
cross-encoder/ms-marco-MiniLM-L-6-v2
Top-30 Candidates ⟶ Top-5 Passages
Generation, Citation Grounding & Reliability Subsystem
11. Hierarchical Context Recovery
Hydrates 270 full Parent Articles & Cases from parent_store.json (Deduplicated)
12. Evidence-Grounded Generator
Gemini API / Offline Legal Fallback
Strict Markdown synthesis with inline [Doc: id]
13. Structural Citation Validator
Deterministic existence & provenance check
Filters unretrieved / fabricated authorities
14. Explainable 6-Signal Confidence
$C = \sum_{i=1}^{6} \omega_i s_i$ (Retrieval, Agreement, Citation...)
15. Safe Multi-Stage Abstention Gate
Threshold $C \geq 0.30$ —
Refuses out-of-scope queries
Verified Output & 10-Stage Inspector
Answer Badges, Entity Pills, Provenance Links, Diagnostic Trace